

AI Website Generator - Software Design Document (Version 1.0)

1. Project Overview

Project Name

AI Website Generator

Goal

Build an AI-powered platform where users can describe a website in natural language, and the system generates a complete React application with a live preview.

Example:

User Input

Build a modern coffee shop website with dark mode and smooth animations.

Output

- React Project
- Tailwind CSS
- Responsive Layout
- Multiple Pages
- Live Preview
- Editable Project

2. Problem Statement

Traditional website development requires:

- UI Design
- HTML/CSS
- React Development
- Responsive Design
- Testing

This project aims to automate these tasks using AI.

3. Objectives

The system should:

- Accept user prompts.
- Understand website requirements.
- Generate a project plan.
- Create React components.
- Build a complete folder structure.
- Install dependencies.
- Launch a development server.
- Display a live preview.
- Allow further edits through additional prompts.

4. Technology Stack

Frontend

- React
 - Vite
 - Tailwind CSS
 - React Router
-

Backend

- FastAPI
 - Python
-

AI

- OpenAI API (or another LLM)
-

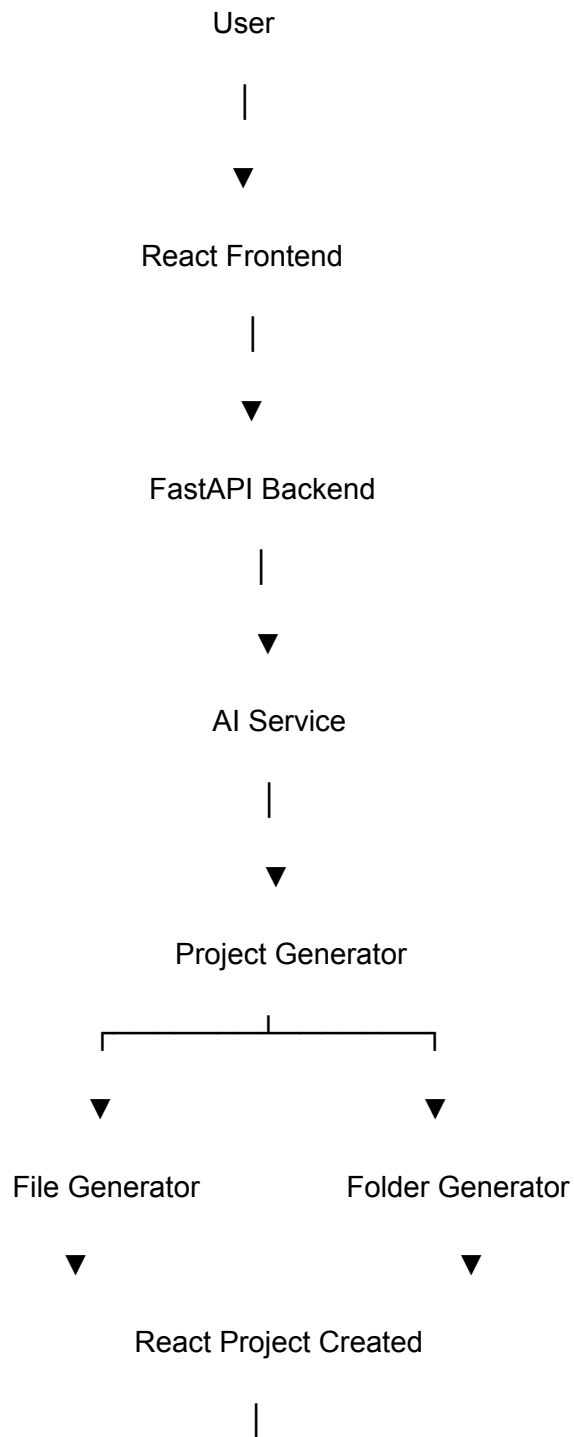
Development Tools

- Git

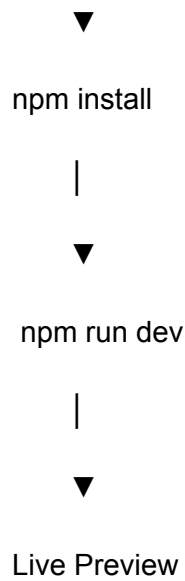
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- VS Code
- npm

5. System Architecture



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6. User Flow

Step 1

User enters:

Build a travel agency website.

Step 2

Frontend sends

```
{
  "prompt": "Build a travel agency website."
}
```

Step 3

Backend receives prompt.

Step 4

AI analyzes the request.

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Step 5

AI creates a project plan.

Step 6

Backend generates project folders.

Step 7

AI generates React files.

Step 8

Python writes files.

Step 9

Dependencies are installed.

Step 10

Development server starts.

Step 11

Website preview is shown.

7. AI Pipeline

Prompt



Prompt Analyzer



Requirement Extractor



Project Planner



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Folder Planner



Component Generator



CSS Generator



Route Generator



API Generator



Validation



File Writer



Preview

8. Prompt Analyzer

Example Prompt

Build a modern portfolio website.

Extract:

Website Type:

Portfolio

Theme:

Modern

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Color :

Dark

Framework :

React

Styling :

Tailwind

Pages :

Home

About

Projects

Contact

Components :

Navbar

Hero

Skills

Projects

Footer

9. Project Planner

Create:

src/

components/

pages/

hooks/

assets/

services/

App.jsx

main.jsx

10. Component Generator

Generate :

Navbar.jsx

Hero.jsx

Footer.jsx

Button.jsx

Card.jsx

11. Styling Generator

Choose

- Tailwind
- CSS Modules
- Styled Components

Generate corresponding styles.

12. Routing Generator

Create:

/

About

Projects

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Contact

using React Router.

13. Validation

Check:

- JSX syntax
- Missing imports
- Duplicate components
- Invalid routes
- Build errors

14. File Generator

Python creates

mkdir components

mkdir pages

write Navbar.jsx

write Hero.jsx

write App.jsx

15. Dependency Manager

Install

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React

React Router

Tailwind

Axios

Framer Motion

16. Preview Engine

Run

npm install

↓

npm run dev

↓

localhost:5173

17. Edit Mode

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User:

Change the navbar to glassmorphism.

System:

- Read current files.
- Update only the navbar.
- Keep the rest of the project unchanged.

18. Future Features

- Multi-page websites
- Next.js support
- Backend generation
- Database integration
- Authentication
- AI image generation
- Deployment to Vercel
- Deployment to Netlify
- SEO optimization
- Performance optimization

19. Folder Structure

AI-Website-Generator/

frontend/

backend/

generated_projects/

templates/

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prompts/

logs/

uploads/

config/

docs/

README.md

20. Development Roadmap

Phase 1

- Prompt input
- React frontend
- FastAPI backend
- Connect to an LLM
- Generate a single React component

Phase 2

- Generate a complete React project
- Create folder structure
- Write files
- Install dependencies
- Run and preview the app

Phase 3

- Support editing existing projects
- Validate generated code

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- Add multiple frameworks
- Improve prompt understanding
- Enable deployment